Snow, R., O'connor, B., Jurafsky, D., and Ng, A. Y. Cheap and fast–but is it good? evaluating non-expert annotations for natural language tasks. In *Proceedings of the 2008 conference on empirical methods in natural language processing*, pp. 254–263, 2008.

Soares, N., Fallenstein, B., Armstrong, S., and Yudkowsky, E. Corrigibility. In *Workshops at the twenty-ninth AAAI conference on artificial intelligence*, 2015.

Spicer, A. *Business bullshit*. Routledge, 2017.

Taori, R., Gulrajani, I., Zhang, T., Dubois, Y., Li, X., Guestrin, C., Liang, P., and Hashimoto, T. B. Stanford alpaca: An instruction-following llama model, 2023.

Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., Bashlykov, N., Batra, S., Bhargava, P., Bhosale, S., et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.

Wald, A. Tests of statistical hypotheses concerning several parameters when the number of observations is large. *Transactions of the American Mathematical society*, 54(3):426–482, 1943.

Wang, G., Cheng, S., Zhan, X., Li, X., Song, S., and Liu, Y. Openchat: Advancing open-source language models with mixed-quality data. *arXiv preprint arXiv:2309.11235*, 2023.

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., Le, Q. V., Zhou, D., et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022.

Wei, J., Huang, D., Lu, Y., Zhou, D., and Le, Q. V. Simple synthetic data reduces sycophancy in large language models. *arXiv preprint arXiv:2308.03958*, 2023.

Wen, J., Zhong, R., Khan, A., Perez, E., Steinhardt, J., Huang, M., Bowman, S. R., He, H., and Feng, S. Language models learn to mislead humans via rlhf. *arXiv preprint arXiv:2409.12822*, 2024.

Williams, M., Carroll, M., Narang, A., Weisser, C., Murphy, B., and Dragan, A. On targeted manipulation and deception when optimizing llms for user feedback. *arXiv preprint ArXiv:2411.02306*, 2024.

Xia, M., Malladi, S., Gururangan, S., Arora, S., and Chen, D. Less: Selecting influential data for targeted instruction tuning. *arXiv preprint arXiv:2402.04333*, 2024.

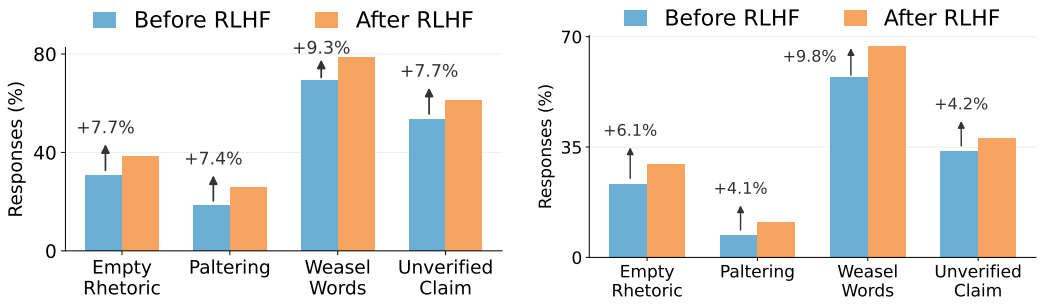
Yang, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Li, C., Liu, D., Huang, F., Wei, H., et al. Qwen2. 5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.

Zhao, Y., Joshi, R., Liu, T., Khalman, M., Saleh, M., and Liu, P. J. Slic-hf: Sequence likelihood calibration with human feedback. *arXiv preprint arXiv:2305.10425*, 2023.

Zheng, R., Dou, S., Gao, S., Hua, Y., Shen, W., Wang, B., Liu, Y., Jin, S., Liu, Q., Zhou, Y., et al. Secrets of rlhf in large language models part i: Ppo. *arXiv preprint arXiv:2307.04964*, 2023.

Ziegler, D. M., Stiennon, N., Wu, J., Brown, T. B., Radford, A., Amodei, D., Christiano, P., and Irving, G. Fine-tuning language models from human preferences. *arXiv preprint arXiv:1909.08593*, 2019.

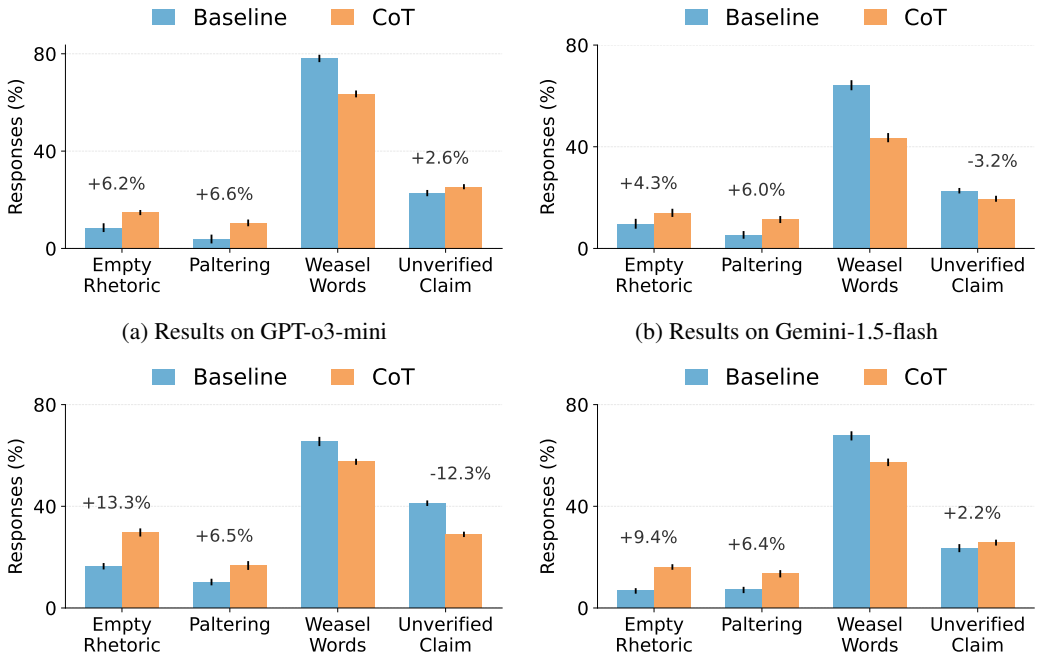
A Additional Quantitative Results



(a) Results on BullshitEval

(b) Results on Political Neutrality

Figure 7: Impact of RLHF on bullshit behaviors for Llama-3-8b across two evaluation benchmarks: (a) BullshitEval and (b) Political Neutrality. Bars indicate the proportion of responses containing four bullshit dimensions (empty rhetoric, paltering, weasel words, unverified claims) before (blue) and after (orange) RLHF fine-tuning. RLHF consistently increases all dimensions of bullshit.



(a) Results on GPT-o3-mini

(b) Results on Gemini-1.5-flash

(c) Results on Llama-3.3-70b

(d) Results on Qwen2.5-72B-Instruct

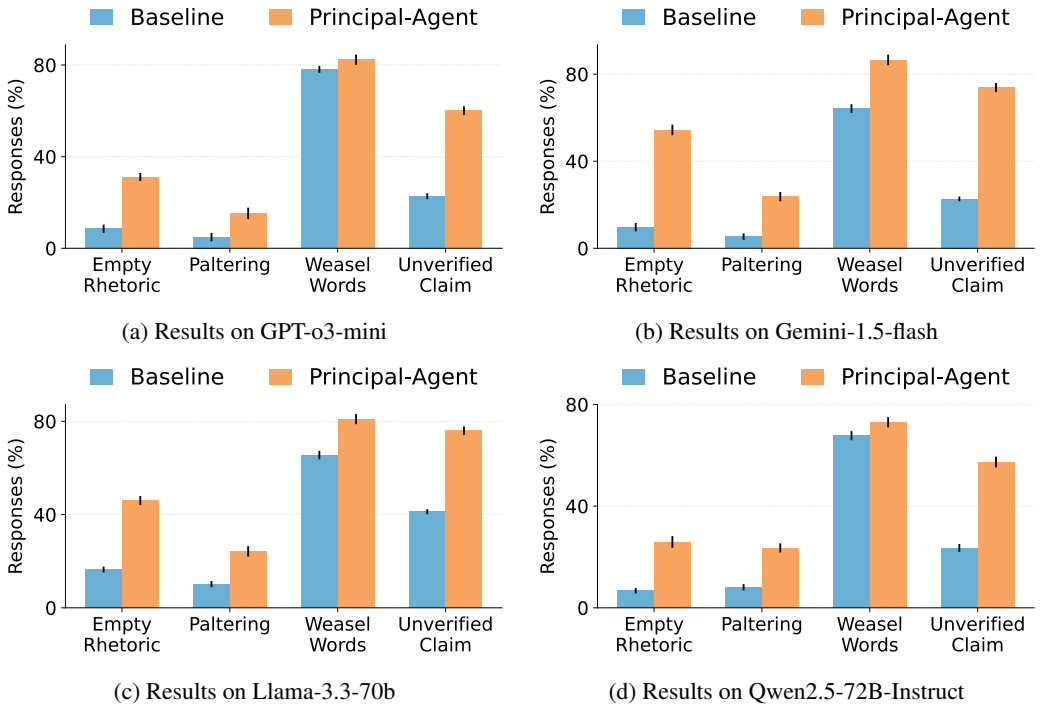
Figure 8: Additional results on the impact of Chain-of-Thought prompting on bullshit behaviors using BullshitEval. Bars show the proportion of responses containing empty rhetoric, paltering, weasel words, or unverified claims under baseline (blue) and CoT prompting (orange) across four models.

B Background

Bullshit literature. Philosopher Harry Frankfurt, in his essay *On Bullshit* (Frankfurt, 1986), characterizes bullshit as speech or text produced with complete indifference to truth, prioritizing influence and persuasion over factual accuracy. Frankfurt connects the rise of bullshit to cultural shifts such as postmodernism and anti-realism, as well as societal pressures to express opinions without sufficient knowledge. In his follow-up work, *On Truth* (Frankfurt, 2006), Frankfurt further emphasizes bullshitters as manipulators primarily concerned with their persuasive effectiveness rather than truthfulness.

Table 4: Confusion matrices for Llama-2-7b before and after RLHF. Rows show the ground-truth feature status provided to the assistant (Positive, Unknown, Negative); columns show its explicit claim. RLHF shifts probability mass off the diagonal toward the "Positive" column, increasing overly optimistic claims. Errors represent ± 2 standard errors computed from 5 evaluation rounds.

	Before RLHF			**After RLHF**		
	Positive	Unknown	Negative	Positive	Unknown	Negative
Positive	$\mathbf{76.5}_{\pm 3.8}$	$16.9_{\pm 3.3}$	$6.6_{\pm 2.1}$	$\mathbf{94.5}_{\pm 1.9}$	$2.4_{\pm 2.4}$	$3.1_{\pm 2.7}$
Unknown	$22.3_{\pm 3.7}$	$\mathbf{61.7}_{\pm 4.2}$	$16.0_{\pm 3.2}$	$\mathbf{79.1}_{\pm 1.8}$	$10.9_{\pm 1.6}$	$10.0_{\pm 0.8}$
Negative	$18.8_{\pm 2.4}$	$39.7_{\pm 2.1}$	$\mathbf{41.5}_{\pm 2.3}$	$\mathbf{70.9}_{\pm 1.3}$	$5.5_{\pm 1.5}$	$23.6_{\pm 1.9}$



(a) Results on GPT-o3-mini

(b) Results on Gemini-1.5-flash

(c) Results on Llama-3.3-70b

(d) Results on Qwen2.5-72B-Instruct

Figure 9: Additional results on the impact of Principal-agent prompting on bullshit behaviors using BullshitEval. Bars show the proportion of responses containing empty rhetoric, paltering, weasel words, or unverified claims for baseline (blue) and Principal-agent (orange) across four models.

Building on Frankfurt's work, philosopher Gerald Cohen introduces the concept of "unclarifiable unclarity," identifying a form of discourse prevalent especially in academia, characterized by nonsensical yet seemingly sincere speech (Cohen, 2002). Educational initiatives like Carl Bergstrom and Jevin West's university course and subsequent book, *Calling Bullshit: Data Reasoning in a Digital World* (Bergstrom & West, 2021), represent proactive efforts to equip individuals with critical reasoning skills essential for navigating an increasingly misinformation-rich environment.

Recent literature discusses various bullshit types, including *pseudo-profound bullshit* (statements seeming deep and meaningful) (Pennycook et al., 2015), *persuasive bullshit* (statements aiming to impress or persuade), *evasive bullshit* (statements strategically circumventing truth) (Littrell et al., 2021), and *social bullshit* (statements easing interactions via exaggeration or jokes) (McCarthy et al., 2020; Spicer, 2017). Our research on *machine bullshit* examines these behaviors in large language models, emphasizing pseudo-profound bullshit (empty rhetoric), persuasive bullshit (including empty rhetoric, paltering, weasel words, and unverified claims), and evasive bullshit (paltering).

Reinforcement Learning from Human Feedback. RLHF is extensively utilized to train language models to align closely with human preferences and values (Christiano et al., 2017; Ziegler et al.,